FINITE TRIPLE COVERS, CONDUCTORS, AND RIGIDIFIED MODULI OF CUBIC-FRAME KELLER MAPS

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Abstract. For every coprime admissible cubic-frame pair (A, B), the projectivized inverse cubic defines a smooth finite-flat Gorenstein triple cover of the target. Its Tschirnhausen bundle is canonically trivialized by the marked root coordinate, and the original Keller source is obtained by deleting the ramification divisor and the unmarked infinity sections. We prove a global conductor formula for the normalization of the discriminant component. If

$$H(c,t) = 3A(c)t + B(c),$$

then the conductor on the normalized discriminant is exactly (H^2); on the discriminant itself it is generated by

$$B^2 - 3Ab, \qquad 18Aa + Bb.$$

Thus the nonnormality is transversely the ordinary cusp $\mathbb{C}[H^2, H^3] \subset \mathbb{C}[H]$, uniformly along the triple-root curve.

We then formulate a moduli statement at the correct level of rigidity. The smooth Deligne–Mumford quotient $[\mathcal{U}_N/\mathbb{G}_m]$ is the actual stack of normalized, Tschirnhausen-framed cubic completions of boundary length N; its inertia is the finite scaling stabilizer of the decorated infinity scheme. The forgetful map to polynomial left–right orbits is bijective on geometric isomorphism classes. It is not the unrigidified left–right moduli stack: every Keller map has an infinite-dimensional space of first-order left–right automorphisms, and stabilization adds the entire automorphism group of the inert affine factor. We isolate the remaining ordinary automorphism problem in an exact sequence whose kernel consists of hidden automorphisms acting trivially on the conductor-decorated boundary.

1. Cubic frames and their projective completion

Let $(A, B) \in \mathbb{C}[c]^2$ be *admissible*:

$$A(0) = 0, \quad A'(0) = 1, \qquad B(0) = -2, \quad B'(0) = -2A''(0), \tag{1}$$

and assume throughout that

$$\gcd(A, B) = 1. \tag{2}$$

The cubic-frame construction produces a polynomial Keller map

$$G_{A,B} = (a, b, c) : \mathbb{A}^3_{x,y,z} \longrightarrow \mathbb{A}^3_{a,b,c}$$

with $\det DG_{A,B} = -2$ and generic degree three. Its inverse polynomial is

$$P_{a,b,c}(T) = A(c)T^3 + B(c)T^2 + bT - 2a. \tag{3}$$

Put $Y = \mathbb{A}^3_{a,b,c}$, let $p : Y \times \mathbb{P}^1_{[U:V]} \to Y$, and homogenize (3):

$$F_{A,B} = A(c)U^3 + B(c)U^2V + bUV^2 - 2aV^3. \tag{4}$$

Define

$$\overline{X}_{A,B} = V(F_{A,B}) \subset Y \times \mathbb{P}^1, \qquad \pi = p|_{\overline{X}_{A,B}}. \tag{5}$$

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Theorem 1.1 (Finite-flat triple completion). *The morphism*

$$\pi : \overline{X}_{A,B} \longrightarrow Y$$

is finite flat of degree three. The total space $\overline{X}_{A,B}$ is smooth, hence it is the normalization of Y in the function field extension defined by $G_{A,B}$. It is a finite-flat Gorenstein triple cover. Moreover, if

$$\mathcal{E} = \left(\pi_* \mathcal{O}_{\overline{X}_{A,B}} / \mathcal{O}_Y\right)^\vee$$

is its Tschirnhausen bundle, then

$$\mathcal{E} \simeq \mathcal{O}_Y^{\oplus 2}. \tag{6}$$

The homogeneous root coordinates $[U : V]$ give the corresponding Tschirnhausen framing, and (4) is the associated framed binary cubic.

Proof. Condition (2) implies that the four coefficients of (4) never vanish simultaneously on a fiber of p. Thus $\overline{X}_{A,B}$ contains no complete fiber of the $\mathbb{P}^1$-bundle. Every geometric fiber is an effective divisor of degree three on $\mathbb{P}^1$, hence is zero-dimensional of length three. The morphism π is proper and quasi-finite, therefore finite.

The defining section gives the exact sequence

$$0 \longrightarrow \mathcal{O}_{Y\times\mathbb{P}^1}(-3) \xrightarrow{F_{A,B}} \mathcal{O}_{Y\times\mathbb{P}^1} \longrightarrow \mathcal{O}_{\overline{X}_{A,B}} \longrightarrow 0.$$

Pushing forward by p, using

$$p_*\mathcal{O}(-3) = 0, \qquad R^1 p_* \mathcal{O} = 0, \qquad R^1 p_* \mathcal{O}(-3) \simeq \mathcal{O}_Y^{\oplus 2},$$

gives an exact sequence

$$0 \longrightarrow \mathcal{O}_Y \longrightarrow \pi_* \mathcal{O}_{\overline{X}_{A,B}} \longrightarrow \mathcal{O}_Y^{\oplus 2} \longrightarrow 0. \tag{7}$$

Hence $\pi_* \mathcal{O}_{\overline{X}_{A,B}}$ is locally free of rank three, so π is finite flat of degree three, and (6) follows. Since $\overline{X}_{A,B}$ is a relative Cartier divisor in the smooth $\mathbb{P}^1$-bundle, the finite-flat cover is Gorenstein.

It remains to check smoothness. On the chart $V \neq 0$, the equation is

$$A(c)T^3 + B(c)T^2 + bT - 2a = 0,$$

and its derivative with respect to a is -2. On the chart $U \neq 0$, with $s = V/U$, the equation is

$$A(c) + B(c)s + bs^2 - 2as^3 = 0.$$

At a point with $s \neq 0$, the derivative with respect to a is $-2s^3 \neq 0$. At a point with $s = 0$, one has $A(c) = 0$, and (2) gives $B(c) \neq 0$; the derivative with respect to s is then nonzero. Thus the total space is smooth and therefore normal. Since it is finite over Y with the same function field as the affine map, it is the normalization of Y in that extension. □

Remark 1.2 (Canonical nature of the completion). Because $\overline{X}_{A,B}$ is the normalization of the target in the generic cubic extension, every left–right equivalence of cubic-frame maps extends uniquely to an isomorphism of their finite triple completions over the induced target automorphism. This is the bridge between the affine orbit problem and the geometry of finite covers.

2. The affine source and the deleted boundary

Near infinity, set $s = V/U$. The equation becomes

$$A(c) + B(c)s + bs^2 - 2as^3 = 0. \tag{8}$$

Let

$$Q_A = A/c, \qquad Z_A = \operatorname{Spec} \mathbb{C}[c]/(Q_A) \subset \mathbb{G}_m.$$

The branch over $c = 0$ is retained in the affine source; the branches over Z_A are deleted. Because $B|_{Z_A}$ is a unit, every deleted infinity branch is simple in the root direction.

Let $\mathcal{R} \subset \overline{X}_{A,B}$ be the ramification divisor of π, and let $\mathcal{I}_{Z_A}$ be the union, with its scheme structure, of the infinity branches over Z_A.

Proposition 2.1 (Affine marked-root chart)**.** *There is a canonical isomorphism*

$$\mathbb{A}^3_{x,y,z} \simeq \overline{X}_{A,B} \setminus (\mathcal{R} \cup \mathcal{I}_{Z_A}) \,. \tag{9}$$

Under this identification, the finite-root coordinate is $T = y + 1/x$, and a finite simple root reconstructs

$$x = \frac{2}{P'(T)}, \qquad y = T - \frac{P'(T)}{2}, \qquad z = \frac{2x - 3x^2y - c}{x^3}. \tag{10}$$

The infinity branch over $c = 0$ is the divisor $x = 0$ in the affine source.

Proof. On the finite-root chart, (10) gives mutually inverse regular maps precisely where the chosen root is simple. At infinity, (8) and the admissibility conditions give a unique simple branch over $c = 0$; direct expansion of the cubic-frame map identifies it with $x = 0$. Every other simple infinity branch lies over Z_A, and no point of $x = 0$ maps there because $c = 0$ on that divisor. These are exactly the deleted loci. □

Corollary 2.2 (Intrinsic nonproperness boundary)**.** *The reduced nonproperness set is the image of the deleted boundary:*

$$S_{G_{A,B}} = D_{A,B} \cup \bigcup_{r \in |Z_A|} V(c - r),$$

where $D_{A,B}$ is the irreducible cubic-discriminant component.

3. Normalization and the conductor

The cubic discriminant is

$$\Delta_{A,B} = B^2b^2 - 4Ab^3 + 8B^3a - 108A^2a^2 - 36ABab. \tag{11}$$

Let

$$R = \mathbb{C}[a, b, c]/(\Delta_{A,B}), \qquad S = \mathbb{C}[c, t].$$

The normalization map $\nu : \operatorname{Spec} S \to \operatorname{Spec} R = D_{A,B}$ is

$$b = -3At^2 - 2Bt, \qquad a = -At^3 - \frac{1}{2}Bt^2. \tag{12}$$

Put

$$H = 3At + B. \tag{13}$$

The preimage of the singular locus is the irreducible curve $L = V(H)$.

Theorem 3.1 (Global conductor formula)**.** *The conductor of the normalization $R \subset S$ is*

$$\mathfrak{c}_{S/R} = H^2S. \tag{14}$$

As an ideal of R, it is

$$\mathfrak{c}_R = \Big(B^2 - 3Ab,\ 18Aa + Bb\Big) \,. \tag{15}$$

Consequently, the conductor divisor on the normalization is exactly

$$2L. \tag{16}$$

At the generic point of L, the normalization is the ordinary cusp

$$\mathbb{C}(c)[H^2, H^3] \subset \mathbb{C}(c)[H]. \tag{17}$$

In particular, the discriminant component is transversely cuspidal along the triple-root curve, with no additional divisorial nonnormal locus.

Proof. On the normalization one has the exact identities

$$H^2 = B^2 - 3Ab, \qquad 2tH^2 = -18Aa - Bb, \qquad H^3 = 3A(tH^2) + BH^2. \tag{18}$$

The first two show that $H^2, tH^2 \in R$.

A Bezout identity $\rho A + \sigma B = 1$, combined with

$$3At^2 + 2Bt + b = 0, \qquad Bt^2 + 2bt - 6a = 0,$$

gives a monic quadratic equation for t over R. Thus

$$S = R[t] = R + Rt$$

as an R-module. It follows from (18) that

$$H^2 S = RH^2 + R(tH^2) \subset R.$$

Hence H^2S is contained in the conductor, and its contraction to R is generated by the two elements in (15).

For the reverse inclusion, pass to the generic point of L. There A is a unit because $A = H = 0$ would imply $A = B = 0$, contrary to (2). Over $K = \mathbb{C}(c)$, solve for the target coordinates:

$$b = \frac{B^2 - H^2}{3A}, \qquad a = \frac{-B^3 + 3BH^2 - 2H^3}{54A^2}. \tag{19}$$

Therefore

$$R \otimes_{\mathbb{C}[c]} K = K[H^2, H^3], \qquad S \otimes_{\mathbb{C}[c]} K = K[H].$$

The conductor of the cusp ring $K[H^2, H^3] \subset K[H]$ is $H^2K[H]$. Any global conductor element must therefore have H-adic order at least two at the generic point of L. Since H is primitive and linear in t, it is irreducible in the UFD S; hence that element is globally divisible by H^2. This proves (14). The recovery identities give (15), and the local description gives the final assertions. □

Corollary 3.2 (Scheme-theoretic boundary marking)**.** *The triple-root curve is intrinsically recovered from the finite cover as the reduced support of the conductor divisor, while its multiplicity two is intrinsic. Thus every left–right equivalence preserves not merely the reduced curve L, but the doubled divisor $2L$.*

4. The normalized framed-completion stack

Fix $N \geq 1$. Write

$$Q(c) = 1 + q_1 c + \cdots + q_N c^N, \qquad R_0(c) = r_0 + r_1 c + \cdots + r_{N-1} c^{N-1},$$

and set

$$A = cQ, \qquad B = -2 - 4q_1 c + c^2 R_0. \tag{20}$$

Let

$$\mathcal{U}_N = \{(q_1, \ldots, q_N, r_0, \ldots, r_{N-1}) : q_N \operatorname{Res}(Q, B) \neq 0\} \subset \mathbb{A}^{2N}. \tag{21}$$

The multiplicative group acts with weights

$$\lambda \cdot q_i = \lambda^i q_i, \qquad \lambda \cdot r_j = \lambda^{j+2} r_j. \tag{22}$$

Definition 4.1 (Normalized framed completion)**.** A normalized framed cubic completion of length N is a finite-flat Gorenstein triple cover $\pi : X \to \mathbb{A}^3$, together with

(i) a Tschirnhausen framing $\mathcal{E} \simeq \mathcal{O}^{\oplus 2}$, hence a canonical embedding $X \hookrightarrow \mathbb{A}^3 \times \mathbb{P}^1_{[U:V]}$;

(ii) a marked infinity line $[1 : 0]$ and a coordinate c, defined up to common nonzero scaling;

(iii) target coordinates (a, b, c) in which the framed binary cubic has the form (4) with (A, B) given by (20);
(iv) the open affine Keller chart obtained by deleting ramification and the unmarked infinity scheme.

Morphisms preserve all this structure, allowing the common scaling (22).

Theorem 4.2 (Framed-completion moduli)**.** *The stack of normalized framed cubic completions of length N is canonically equivalent to*

$$\mathfrak{B}_N = [\mathcal{U}_N/\mathbb{G}_m]. \tag{23}$$

For a geometric point represented by (Q, R_0), its automorphism group in the framed stack is

$$\mathrm{Aut}_{\mathrm{fr}}(Q, R_0) = \{\lambda \in \mathbb{G}_m : \lambda \cdot (Q, R_0) = (Q, R_0)\}. \tag{24}$$

It is a finite subgroup of μ_N, and it is trivial at a generic point. The stack is smooth Deligne–Mumford of dimension $2N - 1$.

Proof. The finite-flat theorem associates a normalized framed completion to every point of $\mathcal{U}_N$. Conversely, the Tschirnhausen frame and marked root coordinate recover the binary cubic coefficients. The admissibility normalization fixes the constant and first-order terms of A, B, while the polynomial Tschirnhaus gauge has a unique representative with $\deg R_0 < N$. The only remaining frame change is the common scaling of c, whose action is exactly (22). This identifies the framed groupoid with the action groupoid $[\mathcal{U}_N/\mathbb{G}_m]$.

Since $q_N \neq 0$, any stabilizer satisfies $\lambda^N = 1$, proving (24). The parameter space is a smooth open subset of $\mathbb{A}^{2N}$, and the stabilizers are finite and reduced in characteristic zero. Hence the quotient is smooth Deligne–Mumford of dimension $2N - 1$. □

Corollary 4.3 (Monic presentation)**.** *Let $\mathcal{U}_N^{\mathrm{mon}} \subset \mathcal{U}_N$ be the slice $q_N = 1$. Then*

$$\mathfrak{B}_N \simeq [\mathcal{U}_N^{\mathrm{mon}}/\mu_N]. \tag{25}$$

Thus the generic framed completion has trivial automorphism group, and every nontrivial inertia group is an explicit finite scaling symmetry of the decorated infinity scheme.

5. Why this is not the unrigidified left–right stack

The geometric orbit theorem says that the geometric points of $\mathfrak{B}_N$ classify stable left–right classes in the cubic-frame family. Equality of orbit sets does not imply equality of moduli groupoids. There are two unavoidable sources of extra inertia.

Proposition 5.1 (Infinitesimal left–right inertia)**.** *Let $F : \mathbb{A}^n \to \mathbb{A}^n$ be any Keller map over $\mathbb{C}$. The tangent space at the identity of its left–right automorphism functor is canonically isomorphic to the space of all polynomial vector fields on the target:*

$$T_e \,\mathrm{Aut}_{\mathrm{LR}}(F) \simeq \mathrm{Der}_{\mathbb{C}}\, \mathbb{C}[y_1, \ldots, y_n]. \tag{26}$$

In particular, the unrigidified left–right moduli problem has infinite-dimensional infinitesimal inertia at every Keller map and cannot be the finite-dimensional Deligne–Mumford stack $\mathfrak{B}_N$.

Proof. Let V be a polynomial vector field on the target. Since $\det JF$ is a nonzero constant, $(JF)^{-1}$ has polynomial entries. Put

$$W = (JF)^{-1}(V \circ F).$$

Over the dual numbers $\mathbb{C}[\varepsilon]/(\varepsilon^2)$, the maps

$$\Phi = \mathrm{id} + \varepsilon W, \qquad \Psi = \mathrm{id} + \varepsilon V$$

are automorphisms and satisfy

$$F \circ \Phi = \Psi \circ F.$$

Conversely, the first-order equation determines W uniquely from V because JF is invertible. This proves (26). ☐

Proposition 5.2 (Inert stabilization automorphisms)**.** *For every $m > 0$, the automorphism group of $F \times \mathrm{id}_{\mathbb{A}^m}$ contains a diagonal copy of $\mathrm{Aut}(\mathbb{A}^m)$:*

$$\chi \longmapsto (\mathrm{id}_{\mathbb{A}^n} \times \chi, \mathrm{id}_{\mathbb{A}^n} \times \chi) .$$

Therefore a literal stable left–right groupoid has growing inertia even at the level of complex points. Any finite-dimensional stable moduli stack must be rigidified by these inert automorphisms.

Remark 5.3 (Correct interpretation)*.* The stack $\mathfrak{B}_N$ is the actual moduli stack of normalized *framed finite completions*. It is also a finite-dimensional rigidification of the cubic-frame left–right orbit problem and has the correct geometric orbit set. It should not be called the full unrigidified stable left–right stack.

6. The ordinary automorphism problem

Let $\mathrm{Aut}_{\mathrm{LR}}(G_{A,B})$ denote the group of pairs of polynomial automorphisms (Φ, Ψ) satisfying

$$G_{A,B} \circ \Phi = \Psi \circ G_{A,B}.$$

Let

$$\mathrm{Stab}(Z_A, \sigma_{A,B}) = \{u \in \mathbb{G}_m : u(Z_A) = Z_A, \quad u^*\sigma_{A,B} = \sigma_{A,B}\} . \tag{27}$$

Theorem 6.1 (Boundary exact sequence)**.** *Boundary reconstruction defines a natural surjective homomorphism*

$$\rho : \mathrm{Aut}_{\mathrm{LR}}(G_{A,B}) \longrightarrow \mathrm{Stab}(Z_A, \sigma_{A,B}). \tag{28}$$

Hence there is an exact sequence

$$1 \longrightarrow K_{A,B} \longrightarrow \mathrm{Aut}_{\mathrm{LR}}(G_{A,B}) \xrightarrow{\rho} \mathrm{Stab}(Z_A, \sigma_{A,B}) \longrightarrow 1, \tag{29}$$

where $K_{A,B}$ is the group of hidden left–right automorphisms acting trivially on the conductor-decorated boundary.

The projection

$$K_{A,B} \longrightarrow \mathrm{Aut}(\mathbb{A}^3_{a,b,c}), \qquad (\Phi, \Psi) \longmapsto \Psi,$$

is injective. Equivalently, there are no nontrivial deck transformations in the kernel.

Proof. The stable boundary Torelli argument applied to a self-equivalence produces a scaling u preserving the decorated infinity scheme; composition of self-equivalences multiplies these scalings, so this defines the homomorphism ρ. Every stabilizing scaling is realized by the explicit diagonal scaling and polynomial Tschirnhaus gauge, proving surjectivity.

If $(\Phi, \mathrm{id}) \in K_{A,B}$, then Φ induces an automorphism of the function field $\mathbb{C}(x, y, z)$ over $\mathbb{C}(a, b, c)$. The generic cubic has Galois closure S_3, and its degree-three subextension has trivial automorphism group. Hence Φ is the identity. This proves the final assertion. ☐

Proposition 6.2 (Finite-cover reformulation of the hidden kernel)**.** *Every element of $K_{A,B}$ extends uniquely to an automorphism of the finite-flat completion $\overline{X}_{A,B}$ over its target automorphism. Thus $K_{A,B}$ is canonically identified with the subgroup of target automorphisms that*

(i) lift to the normalization $\overline{X}_{A,B}$;
(ii) act trivially on the decorated infinity scheme;
(iii) act trivially on the doubled conductor divisor $2L$ on the normalized discriminant.

For a generic parameter, the visible quotient in (29) *is trivial, so the remaining question is exactly whether this hidden kernel vanishes.*

Proof. The extension is unique by theorem 1.2. The three boundary conditions are precisely the statement that the associated element lies in $\ker\rho$, strengthened scheme-theoretically by the conductor theorem. Conversely, an automorphism of the finite completion satisfying these conditions restricts to the affine marked-root chart and lies in the hidden kernel. ☐

Question 6.3 (Hidden-kernel conjecture)**.** Is $K_{A,B} = 1$ for a generic coprime admissible pair? More strongly, is

$$\mathrm{Aut}_{\mathrm{LR}}(G_{A,B}) \simeq \mathrm{Stab}(Z_A, \sigma_{A,B})$$

for every pair in the unit locus?

7. What has been completed

The finite-cover part of the cubic theory is now closed at the following level.

(1) The projective inverse cubic is a smooth finite-flat Gorenstein triple cover of degree three.
(2) Its Tschirnhausen bundle is trivial and its binary cubic is explicitly framed.
(3) The affine Keller source is the complement of ramification and the unmarked infinity sections.
(4) The discriminant normalization has global conductor $(3At + B)^2$, so its nonnormality is uniformly cuspidal along the intrinsic triple-root curve.
(5) The quotient $[\mathcal{U}_N/\mathbb{G}_m]$ is the actual moduli stack of normalized framed completions, with explicit finite inertia.
(6) The full left–right stack is necessarily larger and infinite-dimensional. Its ordinary complex-point automorphism problem has been reduced to the hidden kernel $K_{A,B}$.

The next sharp target is therefore not another parameter family. It is the hidden-kernel conjecture, preferably first on a Zariski-open generic locus.

Verification boundary. The accompanying exact verifier checks all universal conductor identities, the monic integral equation for the normalization coordinate, the cusp coordinate formulas, the weighted scaling action, and the dual-number left–right identity. Finite flatness, smoothness, Tschirnhausen framing, and the automorphism exact sequence are geometric arguments and are not replaced by symbolic calculation.